Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – 2nd part (Parallel Programming)–February 11th 2022

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Questions 1 (4 points) _____

Questions 2 (4 points) _____

Questions 3 (4 points) _____

Questions 4 (4 points) _____

Questions n. 1

Answer the following questions about OpenMP and briefly explain (without explanation the answer will be considered invalid)

A. OpenMP is an API dedicated to the acceleration of applications in distributed-memory environments: true or false?

B. Describe the functionality of the private, shared, default Data Scope Attribute Clauses

C. How can the search of an element inside an (unbalanced) binary tree be accelerated using OpenMP?

D. The maximum number of threads that can be spawned in OpenMP depends on the underlying hardware architecture: true or false?

Questions n. 2

Answer the following questions about parallel algorithms and parallel programming and briefly explain (without explanation the answer will be considered invalid)

A. List and describe the different kinds of parallelism according to the Flynn taxonomy, together with architectures implementing them when possible.

B. It is not recommendable to mix parallelism technologies since doing so increases the complexity of the compiled binaries: true or false?

C. Describe the map parallel pattern and at least one fusion optimization that can be applied to it.

Questions n. 3

Answer the following questions about MPI and briefly explain (without explanation the answer will be considered invalid)

A. Is the synchronization between processes explicit or implicit?

B. Describe the behavior of MPI_Bcast.

C. What are communicators? What is the advantage of defining multiple communicators in an MPI program?

Questions n. 4

Answer the following questions about PThread and briefly explain (without explanation the answer will be considered invalid)

A. How can a thread communicate the result of its computation after completing its routine?

B. Can PThread be used to implement a fixed-depth parallel pipeline? How should the pipeline stages synchronize their execution?